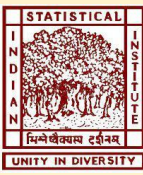


SUMMER INTERNSHIP-2026

DATA VISUALISATION

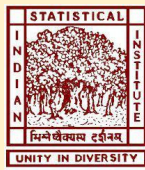
By: Koulika Paul

Date: May 20, 2026



What is Data



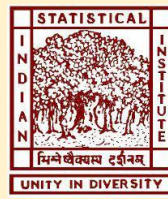


Data refers to raw facts, figures, or unprocessed information that can be collected, measured, analyzed, and used for decision-making or analysis.

Data is a piece of information. Example: Say a student name, school name, age of student, student nationality e.t.c .

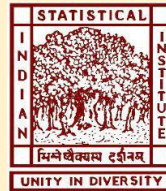
 Name _____ Age _____ Address _____ Tel. _____	 Name _____ Age _____ Address _____ Tel. _____
 Name _____ Age _____ Address _____ Tel. _____	 Name _____ Age _____ Address _____ Tel. _____





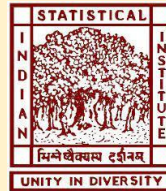
IMPORTANCE OF DATA

- ★ Helps in **decision making**
- ★ Drives **innovation** and **optimization**
- ★ Powers **AI & machine learning**
- ★ Supports **business insights**



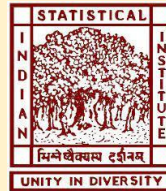
EXPLORATORY DATA ANALYSIS (EDA)

- Process of analyzing and summarizing datasets
- Helps understand data structure and characteristics
- Detects patterns, trends, and anomalies
- Used before statistical modeling or machine learning
- Improves data quality and decision-making



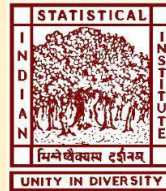
OBJECTIVES OF EDA

- Summarize dataset characteristics
- Identify missing values and outliers
- Understand data distributions
- Detect relationships among variables
- Improve statistical analysis and model accuracy



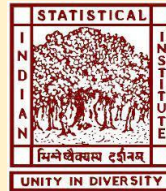
IMPORTANCE OF EDA

- EDA is the foundation of statistical analysis
- Helps understand and clean data
- Improves analytical accuracy
- Essential for machine learning and research
- Good analysis begins with good exploration of data



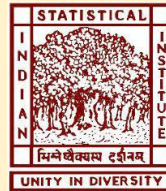
EDA IN STATISTICS

- Descriptive statistics: Mean, Median, Mode
- Measures of variability: Variance, Standard Deviation
- Relationship analysis using Correlation and Covariance
- Helps understand statistical behavior of data



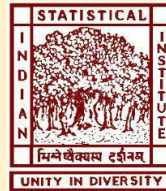
STATISTICAL TECHNIQUES

- Frequency distribution analysis
- Probability distributions
- Correlation analysis
- Visualization-based exploration
- Identifies hidden patterns in data



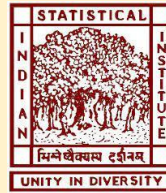
DATA VISUALIZATION IN EDA

- Histograms for distribution analysis
- Boxplots for outlier detection
- Scatterplots for relationship analysis
- Heatmaps for correlation analysis
- Makes interpretation easier



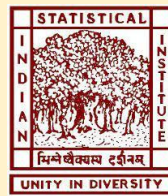
EDA WORKFLOW

- Data Collection
- Data Cleaning
- Data Transformation
- Statistical Summary
- Visualization and Interpretation

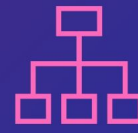
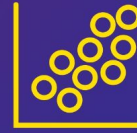
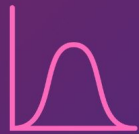
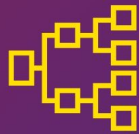


APPLICATIONS

- Business: Sales and customer analysis
- Healthcare: Disease prediction
- Finance: Fraud detection and risk analysis
- Machine Learning: Data preprocessing and feature selection



WHAT IS DATA VISUALIZATION?



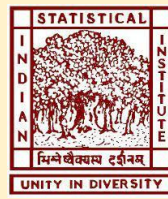


Data Visualization is the graphical representation of information and data using visual elements such as charts, graphs, maps, and dashboards.

It helps transform large and complex datasets into visuals that are easier to understand, interpret, and communicate.

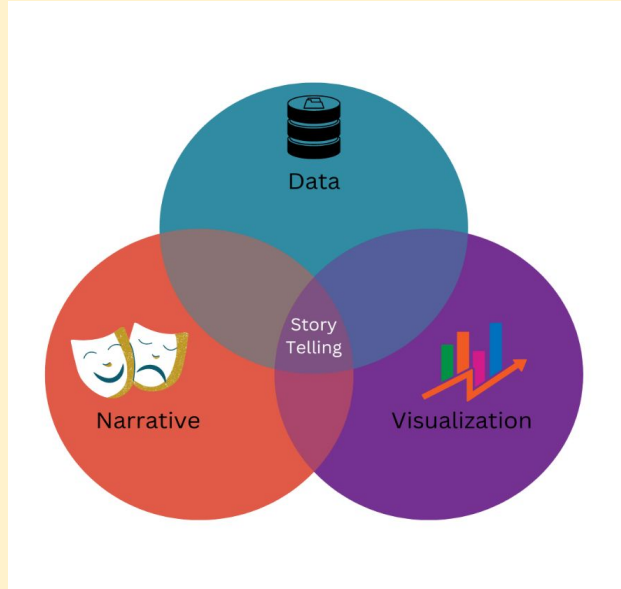
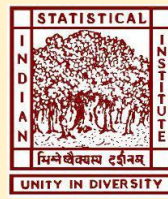
Instead of reading through rows of numbers or text-based data, viewers can use visual cues like bars, lines, and colors to quickly grasp trends, patterns, comparisons, and outliers.

This makes it an essential tool for data analysis, business intelligence, and decision-making.

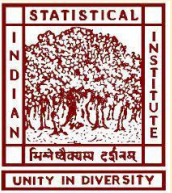


WHY WE NEED DATA VISUALISATION?

- **Enhances Data Storytelling:** Turns raw data into a clear, engaging narrative.
- **Supports Decision Making:** Helps stakeholders make informed, data-driven choices.
- **Aids Problem Solving:** Reveals patterns and outliers on data that point to solutions.
- **Improves Understanding:** Makes complex data easier to grasp at a glance.
- **Identifies Relationships:** Highlights connections and correlations between variables.
- **Enables Comparative Analysis:** Simplifies comparing datasets, categories, or time periods.



DATA VISUALISATION TO DATA STORYTELLING



FLORENCE NIGHTINGALE STORY

Florence Nightingale, also known as the “Lady with the Lamp”, was the nursing administrator of a sprawling British Army hospital network

Many government leaders accepted the loss of common soldiers as inevitable. They wrongly believed, for example, that communicable diseases were caused by unavoidable realities—the weather, bad diet and harsh work conditions.

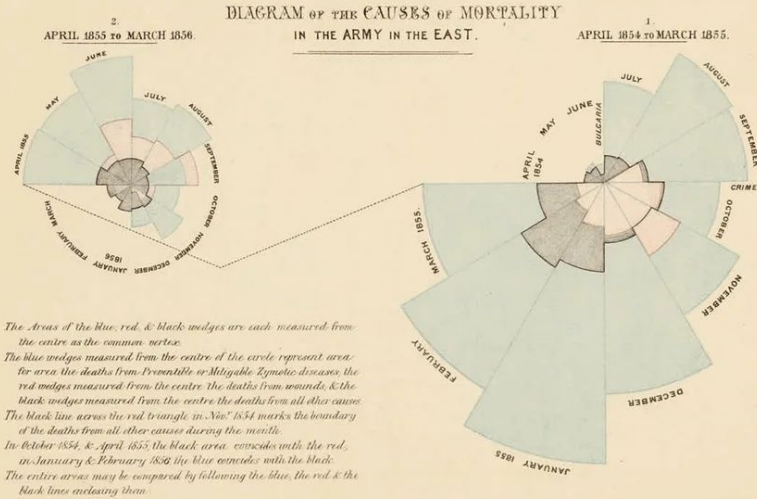
And the poor quality of army data made it impossible to know exactly how soldiers died.

Nightingale's key persuasion tactic was to convey statistics in exciting ways

Instead of building complex arguments that required heavy work from the audience, she focused her narrative lens on specific claims. It was more than data visualization—it was data storytelling.

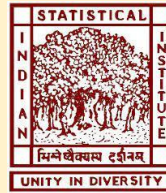
Nightingale's stories showcased how poor sanitation and overcrowding caused unnecessary death.

She showed how, for example, peacetime soldiers living in army barracks died at higher rates than civilian men of similar ages. Her graphics made it impossible to deny the realities represented by the data: army administration needed dramatic reform.



This image represents the mortality of soldiers yearly

- blue area — the deaths caused by preventable, mitigable diseases
- Red area — deaths caused due to wounds
- Black area — deaths caused due to other reasons

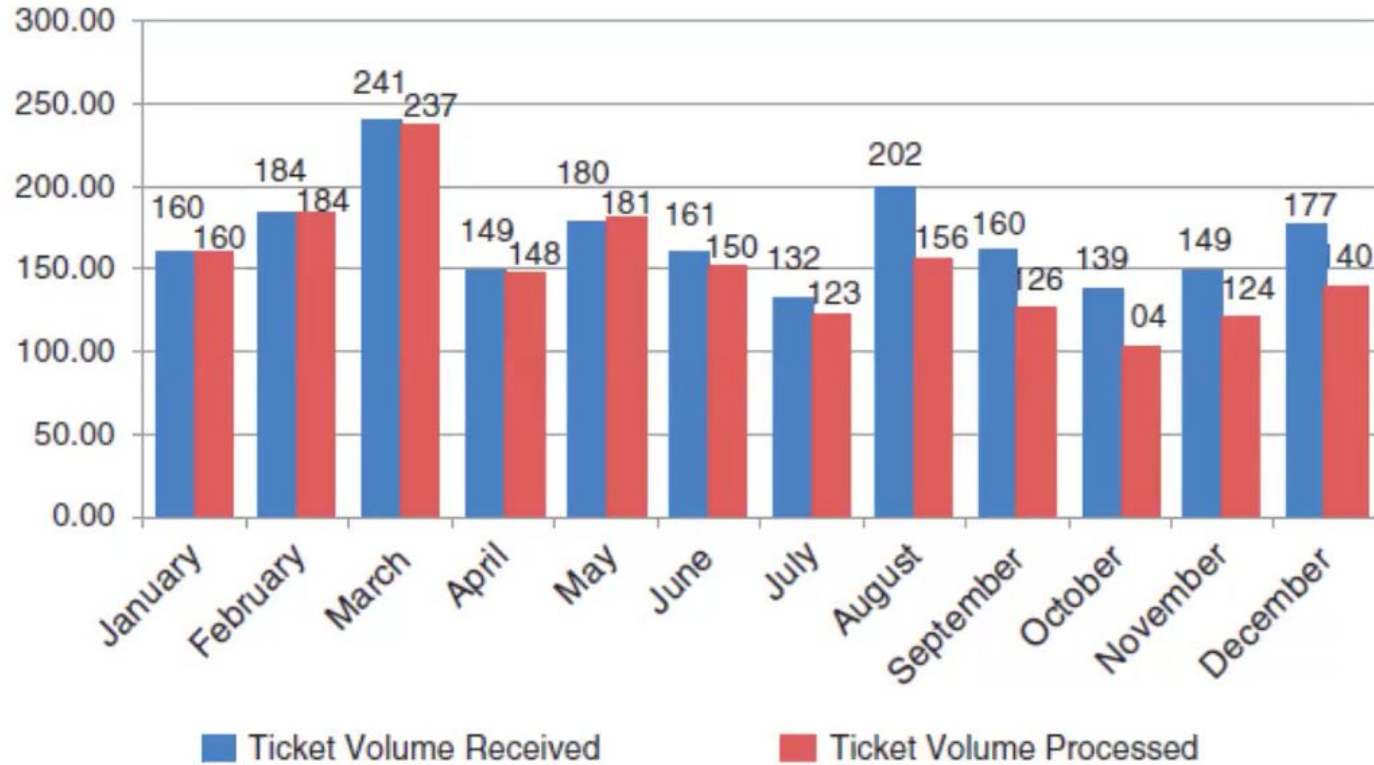


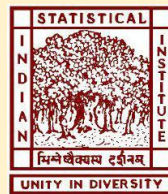
PLANS OF DATA VISUALIZATION

- **Visualisation conveys statistics in more exciting ways.**
- **Define the Purpose:** Know what message or insight you want to convey.
- **Know Your Audience:** Design the visuals according to the knowledge level of your audience (e.g., technical or non-technical).
- **Understand the Data:** Preprocess and analyze the data to understand its format and characteristics.
- **Choose the Right Chart Type :** Choose a visual that best fits the data and the purpose (e.g., bar chart for comparisons, line chart for trends).
- **Simplify and Focus:** Avoid clutter, unnecessary elements and highlight the most important insights.
- **Use Color and Labels Wisely:** Choose colors wisely to differentiate data and make sure all labels are informative and visible.
- **Test and Review:** Review with a colleague or user and refine based on feedback.



Ticket Trend

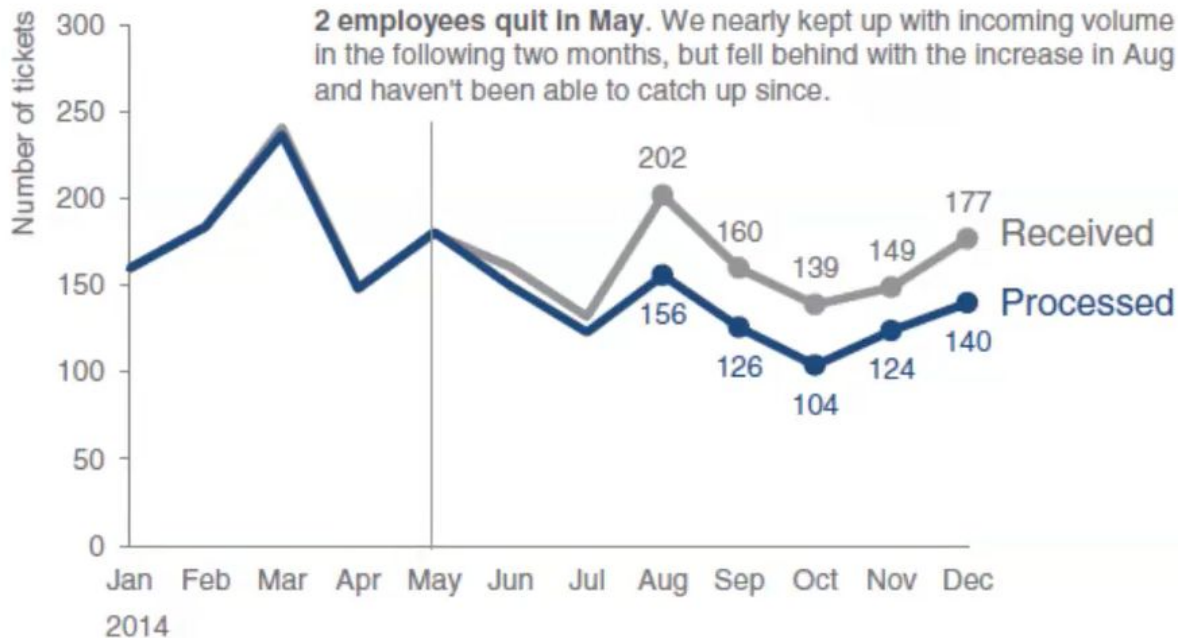


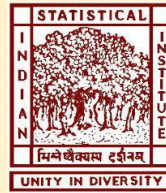


Please approve the hire of 2 FTEs

to backfill those who quit in the past year

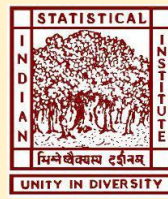
Ticket volume over time





TYPES OF DATA VISUALISATION

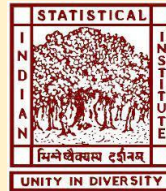
- **Bar Charts:** Used to compare values across categories using rectangular bars.
- **Line Graphs:** Show trends or changes over time by connecting data points with lines.
- **Pie Charts:** Represent proportions or percentages of a whole using circular slices.
- **Histograms:** Display the distribution of continuous data grouped into ranges or bins.
- **Scatter Plots:** Show relationships or correlations between two numerical variables.
- **Heatmaps:** Use color gradients to represent values in a matrix, highlighting intensity or density.
- **Maps (Geo-visualization):** Visualize data across geographical regions using color, markers, or shapes.



TOOLS FOR DATA VISUALISATION

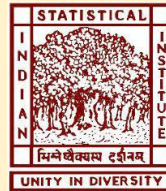
- ❑ **MICROSOFT EXCEL**
- ❑ **MICROSOFT POWER BI**
- ❑ **TABLEAU**
- ❑ **GOOGLE DATA STUDIO**
- ❑ **QLIK SENSE**





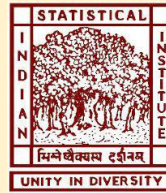
GRAMMAR OF GRAPHICS

*A STRUCTURED APPROACH TO
VISUALISATION*



WHAT IS GRAMMAR OF GRAPHICS?

- The concept was introduced by Leland Wilkinson in his book "The Grammar of Graphics."
- The Grammar of Graphics is a comprehensive framework for creating and understanding data visualizations. It provides a structured approach to constructing graphical displays of data
- Basically, a grammar of graphics follows a layered approach to describe and construct visualizations or graphics in a structured manner.
- It defines a grammar, or set of rules, for building visualizations in a systematic way.
- Understanding this grammar allows for creating complex and customizable graphics.



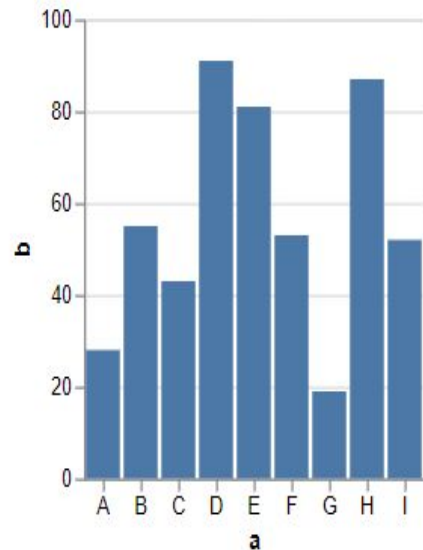
WHY USE GRAMMAR OF GRAPHICS?

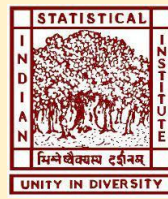
- As we know it is quite difficult to visualize more than 2 dimensional data , so Grammar of graphics is mainly required to visualize multidimensional data, regardless the complexity of the data.
- Grammar of graphics uses several layered components to describe any graphic or visualization.
- It helps in customizing the visuals/graphics.



IMPLEMENTATION OF GRAMMAR OF GRAPHICS VIA VEGALITE

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  "description": "A simple bar chart with embedded data.",
  "data": {
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    ]
  },
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  "encoding": {
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}
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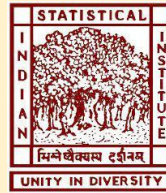
FUNDAMENTALS OF GRAMMAR OF GRAPHICS

. COMPONENTS :

- Components are the basic building blocks that define the elements involved in creating a visualization. They are more about the static elements that make up a graphic.
- Basic elements that builds the set of rules or grammar for the visualisation are components

. LAYERS :

- Layers are a higher-level concept that represents different stages or aspects of the visualization process. They describe how the components are combined and manipulated to create the final visual output
- When values are added to each components for making a visual that represents the layers of the graphics



SUMMARY

- The Grammar of Graphics provides a powerful framework for creating detailed and meaningful visualizations.
- Understanding its components and layers allows for flexibility and precision in data representation.